

CANDIDATE
NAME

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INDEX NUMBER

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BIOLOGY

9648/02

Paper 2 Core Paper

17 September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any one question.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
Section A	80
1	
2	
3	
4	
5	
6	
7	
Section B	20
8 / 9	
Total	100

Section A

Answer **all** the questions in this section.

- 1(a)** In a DNA structure, what do 5' and 3' represent. [2]

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- (b)** The nitrogenous bases of the double helix are paired in specific combinations:

adenine (A) with thymine (T)
guanine (G) with cytosine (C)

It was mainly by trial and error that Watson and Crick arrived at this key feature of DNA. At first, Watson imagined that the bases paired like with like—for example, A with A. But this model was inconsistent with the uniform width of the double helix.

- Explain why the earlier DNA model would have inconsistent width. [3]

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- (c)** DNA replicates semi-conservatively by the two strands separating and forming templates for the synthesis of two new strands.

In addition to the template provided by the DNA, describe the role of two other types of factors that are required for DNA replication. [2]

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- (d) Just as a DNA strand provides a template for making a new complementary strand during DNA replication, it also can serve as a template for transcription.

Fig. 1.1 is a diagrammatic representation of the process of transcription.

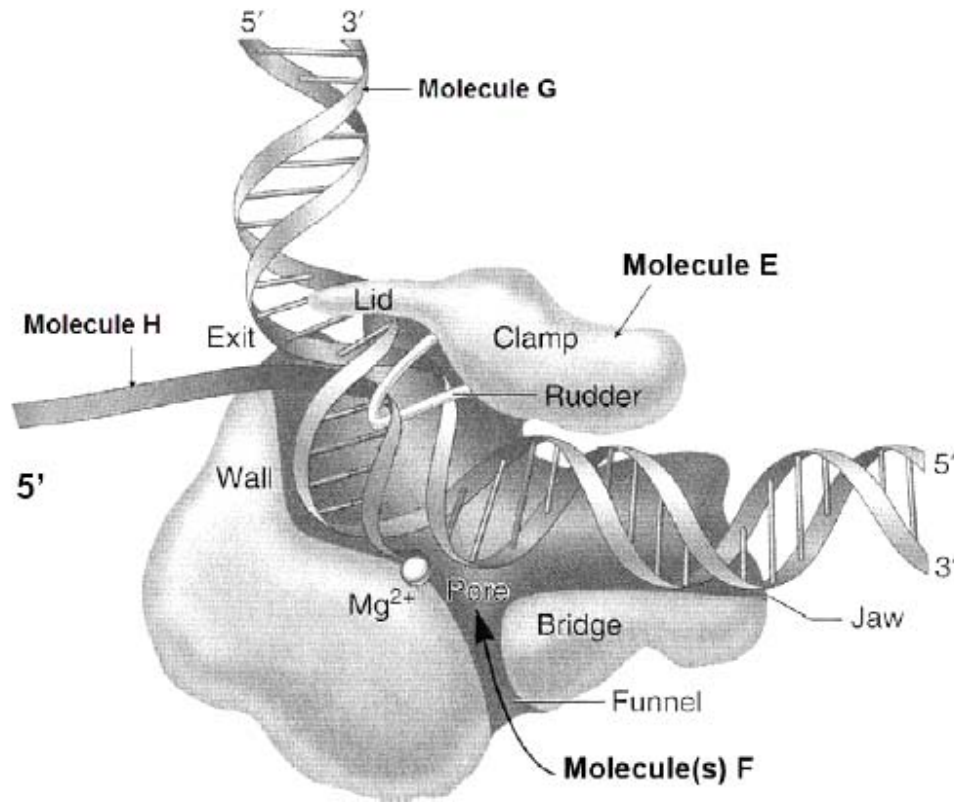


Fig. 1.1

Identify molecule E, F and H.

[1]

E –

F –

H –

- (i) Describe how the structure of molecule E is adapted to its role in transcription.

[3]

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- (ii) Describe one way by which the structure of molecule(s) **F**, which is / are adapted for [1] transcription, differs from that of amino acids, which are adapted for translation.

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[Total : 12]

2 Influenza A viruses are divided into subtypes based on two proteins on the surface of the virus: haemagglutinin (HA) and neuraminidase (NA). The influenza A (H7N9) virus designation of H7N9 identifies it as having HA of the H7 subtype and NA of the N9 subtype.

- (a)** Describe the roles of haemagglutinin (HA) and neuraminidase (NA) in the reproductive cycle of the influenza (H7N9) virus. [2]

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- (b)** **(i)** The influenza genome comprises genes for RNA polymerase. How is the virus RNA polymerase different from the host RNA polymerase? [1]

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- (ii)** Explain the role of the viral RNA polymerase in the reproductive cycle. [3]

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- (c)** H7N9 is an avian influenza virus but it is also able to infect mammals.

Suggest changes that may have occurred to give H7N9 *greater* ability to infect mammals. [2]

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- (d) While the influenza genome comprises 8 distinct linear segments of negative sense ssRNA, bacterial genome consists of chromosome and plasmid. Operons are needed for the regulation of gene expression.

Suggest why operons give bacteria a selective advantage.

[3]

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[Total : 11]

- 3 In normal person, the fragile X mental retardation protein (FMRP), regulates the synthesis of neurone proteins by stopping ribosomal translocation on target mRNAs. Fig 3.1 shows how patients with fragile X syndrome (FXS) have non-functional FMRP, resulting in accelerated synaptic protein synthesis that leads to abnormal synaptic function and intellectual disability.

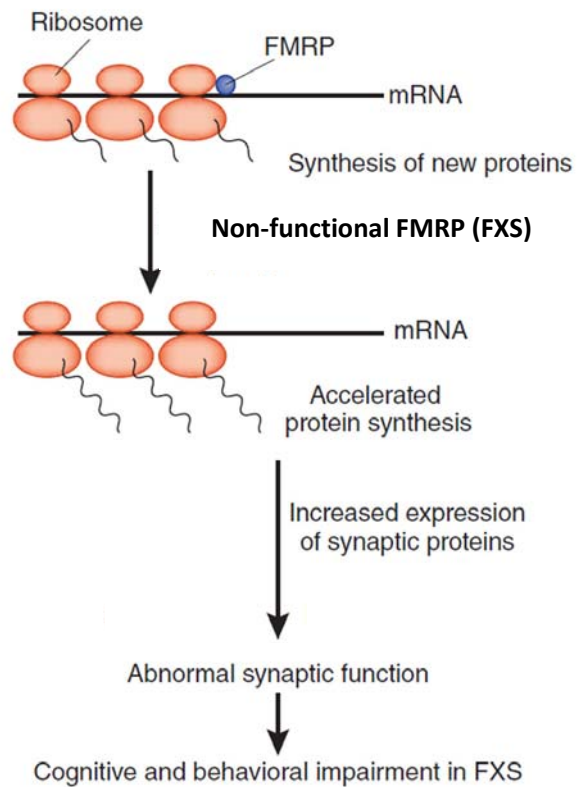


Fig. 3.1

- (a) Describe how polysomes are formed.

[2]

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- (b) (i) State the level of control by FMRP on synaptic protein expression.

[1]

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- (ii)** Describe one other control mechanism of a similar level as **(b)(i)**. [2]

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- (iii)** Explain how FMRP stops the ribosome from translocation. [2]

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- (c)** Initial gene therapy trial uses a short stretch of anti-sense RNA. Explain how this will help to reduce the symptom of FXS. [2]

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- (d)** Synaptic proteins that are folded wrongly are usually degraded.
Suggest the advantages of degrading proteins that are wrongly folded. [2]

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[Total : 11]

- 4 Severe combined immunodeficiency (SCID) results from defects in signaling downstream of cytokine receptors that contain the common cytokine-receptor γ -chain (γ_c). γ_c is a component of the receptors (R) for many interleukin such as interleukin-2 (IL-2) and IL-7.

Fig. 4.1 shows the events in the JAK-STAT cell signaling pathway when cytokine binds and this in turn leads to an activated receptor. Fig. 4.2 shows two causes that could lead to a non-functional receptor.

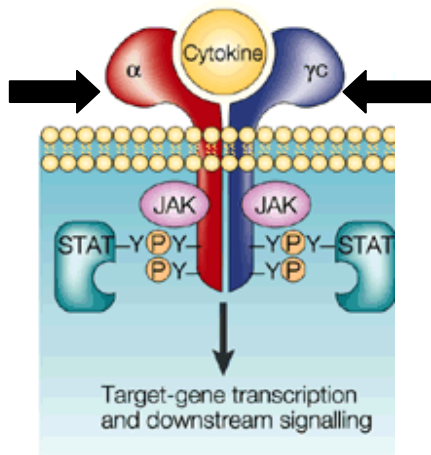


Fig. 4.1

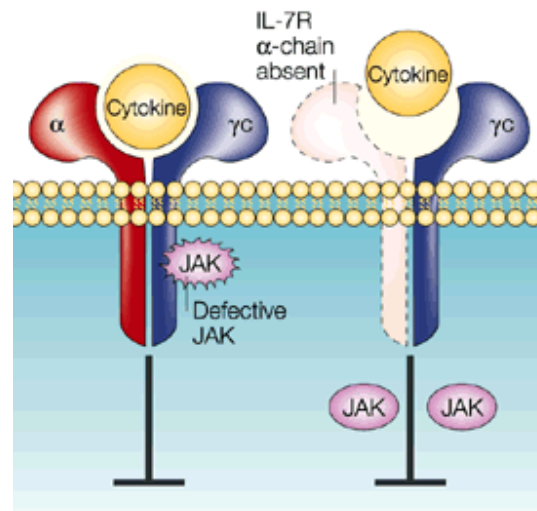


Fig. 4.2

- (a) Explain why transmembrane proteins such as the cytokine receptor are especially suited for a role in cell signalling. [2]

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- (b) With reference to Fig. 4.1, suggest what happens after cytokine binds to the receptor. [2]

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- (c) With reference to Fig. 4.2, explain how a change in the sequence of the DNA nucleotide may affect the signaling downstream of cytokine receptors. [5]

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- (d) Proteins synthesized by ribosomes attached to the endoplasmic reticulum may be embedded in the cell surface membrane. Outline the route taken by such a protein. [3]

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[Total: 12]

- 5 Chickens have a structure called a comb on their heads. The drawings show two types of comb.



Fig. 5.1

The shape of the comb is controlled by two alleles of one gene. The allele for pea comb, **A**, is dominant to the allele for single comb, **a**. The colour of chicken eggs is controlled by two alleles of a different gene. The allele for blue eggs, **B**, is dominant to the allele for white eggs, **b**.

The genes for comb shape and egg colour are situated on the same chromosome.

A farmer crossed a chicken that had pea comb and produced blue eggs with a chicken that had a single comb and produced white eggs. The chicken that had a pea comb and produced blue eggs are the offspring of the cross between pure bred pea comb, blue eggs and pure bred single comb, white eggs.

- (a) Construct a genetic diagram to illustrate the above cross.

[3]

- (b)** The farmer could identify which of the female offspring from this cross would eventually produce blue eggs. Explain how. [2]

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- (c)** However, in certain crosses, the farmer was not able to identify the female offspring which will produce blue eggs. Explain why. [2]

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- (e) In chickens the males are the homogametic sex while the females are the heterogametic sex.

A gene on the X chromosome controls the rate of feather production. The allele for slow feather production, **F**, is dominant to the allele for rapid feather production, **f**.

A farmer made a cross between two chickens with known genotypes. He chose these chickens so that he could tell the sex of the offspring soon after they hatched by looking at their feathers.

Which of the crosses shown in the table did he make? Explain your answer.

Cross	Genotype of male parent	Genotype of female parent
A	$X^F X^F$	$X^f Y$
B	$X^F X^f$	$X^f Y$
C	$X^f X^f$	$X^F Y$
D	$X^F X^f$	$X^F Y$

Fig. 5.3

[2]

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- (f) Female chickens are more likely than male chickens to show recessive sex-linked characteristics. Explain why.

[2]

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[Total: 11]

- 6 Table 6.1 shows the approximate percentage of protein in different membranes.

Table 6.1

Organelle	Membrane	Protein as a percentage of dry weight
Chloroplast	Outer membrane	54
Chloroplast	Inner membrane	70
Mitochondrion	Outer membrane	55
Mitochondrion	Inner membrane	80

- (a) Describe one way in which the distribution of protein in the inner and outer membranes of these two organelles is similar.

Suggest a reason for this similarity.

[2]

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- (b) (i) Atrazine is a widely used herbicide (weedkiller). It binds to a chloroplast protein involved in electron transfer between photosystems.

Explain how atrazine works as an effective herbicide.

[4]

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- (ii) Maize plants are insensitive to atrazine.

The cytoplasm of maize plant cells contains a tripeptide which binds to atrazine. The product is then transported to the vacuole.

Suggest why maize plants are insensitive to atrazine.

[2]

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- (c) At the end of photosynthesis, glyceraldehyde 3-phosphate (3-carbon sugar) is produced by the Calvin cycle, which is subsequently converted into starch for storage. [3]

Discuss the suitability of starch as a storage compound in plants.

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[Total : 11]

- 7 The marine threespine sticklebacks, *Gasterosteus aculeatus* is a freshwater fish living in the lakes of British Columbia, Canada as shown in Fig. 7.1.

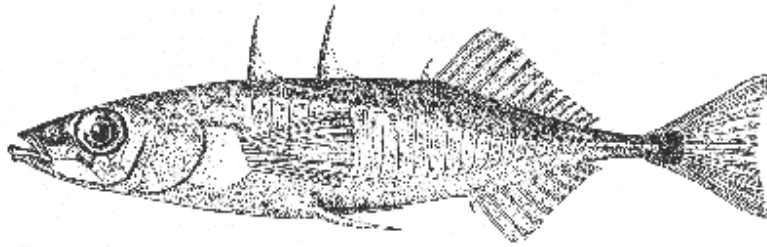


Fig. 7.1

In order to investigate the process of speciation in these populations, three small lakes were studied. Each lake contained two varieties of stickleback: a large, bottom-dwelling variety that fed on invertebrates near the shore and a small, plankton-eating variety that lived in the open water. The probability of breeding between pairs of individuals was measured under laboratory conditions in the following breeding combinations:

- I different varieties from the same lake
- II different varieties from different lakes
- III same variety from different lakes
- IV same variety from the same lake

The data are summarized in Fig. 7.1 below.

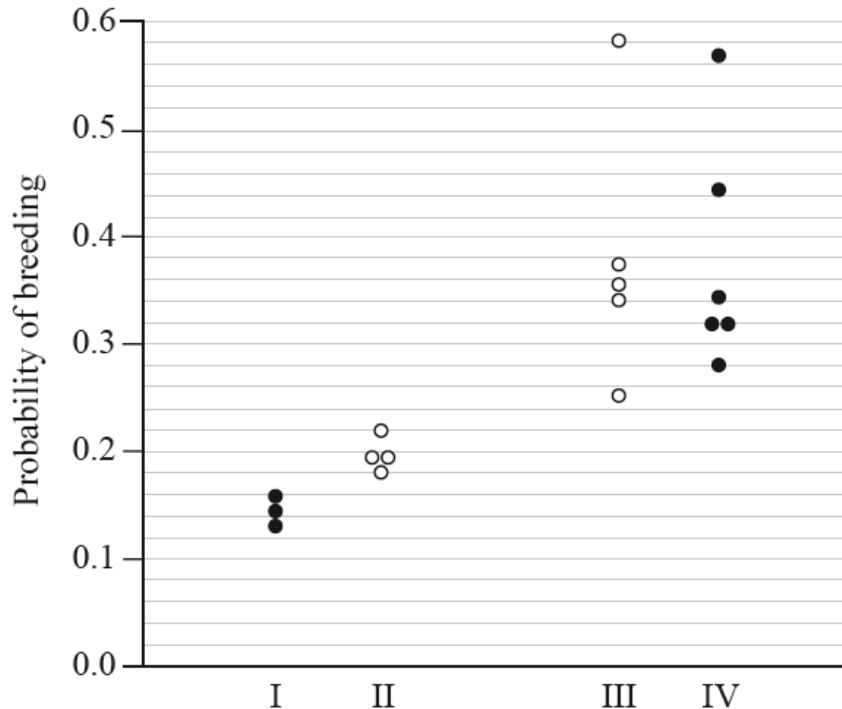


Fig. 7.1

- (a) (i)** Identify the highest and lowest probabilities of breeding for individuals of the same variety from different lakes. [1]

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- (ii)** With reference to Fig. 7.1, describe the differences in probability of breeding between individuals from the same lake. [2]

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- (b)** Scientists concluded that speciation is taking place in these populations. Explain the type of speciation as shown in Fig. 7.1. [4]

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- (c)** With reference to Fig. 7.1, explain why all the individuals are still considered the same species. [2]

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- (d) The freshwater lakes also contain many different types of parasites that infect the marine threespine sticklebacks. Suggest why these parasites help to speed up speciation of the marine threespine sticklebacks. [3]

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[Total : 12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section (a), (b) etc., as indicated in the question.

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|----------|---|------|
| 8 | (a) Explain how the positive and negative control of the lac operon affect bacteria growth. | [10] |
| | (b) Describe how regulation of blood glucose level in humans involves cAMP. | [5] |
| | (c) Explain the significance of the steps in glycolysis. | [5] |
| 9 | (a) Describe the differences between cellulose synthase and cellulose. | [5] |
| | (b) With reference to photosynthesis and cellular respiration, explain the importance of compartmentalization within a plant cell. | [10] |
| | (c) Explain the role of meiosis in natural selection. | [5] |